# Similarity Error Analysis TEACHER KEY



Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

## 1. Setting up a proportion

**ORIGINAL PROBLEM**

△ABC ~ △DEF with AB = 6, DE = 9 and BC = 8. Find EF.

**INCORRECT WORKED SOLUTION**

$\frac{AB}{DE} = \frac{EF}{BC}$

$\frac{6}{9} = \frac{x}{8}$

$9x = 48$

$x = \frac{16}{3}$

**A. EXPLAIN THE MISTAKE (1 POINT)**

Corresponding sides must be in the SAME positions on both sides of the proportion: $\frac{AB}{DE} = \frac{BC}{EF}$. The wrong work swapped BC and EF.

**B. SHOW CORRECT WORK (1 POINT)**

$\frac{AB}{DE} = \frac{BC}{EF}$

$\frac{6}{9} = \frac{8}{x}$

$6x = 72$

$x = 12$

## 2. Area and scale factor

**ORIGINAL PROBLEM**

Two similar figures have scale factor 3. The smaller has area 10. Find the larger area.

**INCORRECT WORKED SOLUTION**

Area = 10 · 3

Area = 30

**A. EXPLAIN THE MISTAKE (1 POINT)**

Areas scale by the SQUARE of the scale factor: $3^2 = 9$. Area = 10 · 9 = 90.

**B. SHOW CORRECT WORK (1 POINT)**

Area ratio = $3^2 = 9$

Area = 10 · 9 = 90

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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## 3. Perimeters of similar figures

**ORIGINAL PROBLEM**

Similar polygons have scale factor 2 : 5. The smaller has perimeter 24. Find the larger perimeter.

**INCORRECT WORKED SOLUTION**

$\frac{2}{5} = \frac{x}{24}$

$5x = 48$

$x = 9.6$

**A. EXPLAIN THE MISTAKE (1 POINT)**

The larger polygon must have a perimeter BIGGER than 24. Small-to-large must match on both sides: $\frac{2}{5} = \frac{24}{x}$.

**B. SHOW CORRECT WORK (1 POINT)**

$\frac{2}{5} = \frac{24}{x}$

$2x = 120$

$x = 60$

## 4. AA similarity

**ORIGINAL PROBLEM**

△ABC has angles of 50° and 60°. △DEF has angles of 60° and 80°. Are they similar?

**INCORRECT WORKED SOLUTION**

Both triangles have a 60° angle.

AA similarity applies.

The triangles are similar.

**A. EXPLAIN THE MISTAKE (1 POINT)**

AA needs TWO pairs of congruent angles. △ABC: 50°, 60°, 70°. △DEF: 60°, 80°, 40°. Only one pair matches.

**B. SHOW CORRECT WORK (1 POINT)**

△ABC: 50°, 60°, 70°

△DEF: 60°, 80°, 40°

Only one pair of angles matches.

The triangles are not similar.

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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## 5. Indirect measurement

**ORIGINAL PROBLEM**

A 6-ft person casts a 4-ft shadow. At the same time, a tree casts a 20-ft shadow. Find the tree's height.

**INCORRECT WORKED SOLUTION**

$\frac{6}{4} = \frac{20}{x}$

$6x = 80$

$x = \frac{40}{3} \approx 13.3$ ft

**A. EXPLAIN THE MISTAKE (1 POINT)**

Heights must match heights: $\frac{6}{4} = \frac{x}{20}$. The wrong proportion put the tree's shadow (20) in the height position.

**B. SHOW CORRECT WORK (1 POINT)**

$\frac{\text{person height}}{\text{person shadow}} = \frac{\text{tree height}}{\text{tree shadow}}$

$\frac{6}{4} = \frac{x}{20}$

$4x = 120$

$x = 30$ ft

## 6. Side-splitter theorem

**ORIGINAL PROBLEM**

In △ABC, DE ∥ BC with D on AB and E on AC. AD = 4, DB = 6 and AE = 6. Find EC.

**INCORRECT WORKED SOLUTION**

$\frac{AD}{AB} = \frac{AE}{EC}$

$\frac{4}{10} = \frac{6}{x}$

$4x = 60$

$x = 15$

**A. EXPLAIN THE MISTAKE (1 POINT)**

AB is the WHOLE side (AD + DB) but EC is only a PART of AC. Compare parts to parts: $\frac{AD}{DB} = \frac{AE}{EC}$.

**B. SHOW CORRECT WORK (1 POINT)**

$\frac{AD}{DB} = \frac{AE}{EC}$

$\frac{4}{6} = \frac{6}{x}$

$4x = 36$

$x = 9$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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## 7. Volume and scale factor

**ORIGINAL PROBLEM**

Two similar prisms have scale factor 2. The smaller has volume 12. Find the larger volume.

**INCORRECT WORKED SOLUTION**

Volume = 12 · 2

Volume = 24

**A. EXPLAIN THE MISTAKE (1 POINT)**

Volumes scale by the CUBE of the scale factor: $2^3 = 8$. Volume = 12 · 8 = 96.

**B. SHOW CORRECT WORK (1 POINT)**

Volume ratio = $2^3 = 8$

Volume = 12 · 8 = 96

## 8. Finding a scale factor

**ORIGINAL PROBLEM**

A side of length 5 corresponds to a side of length 15 in a similar figure. Find the scale factor from small to large.

**INCORRECT WORKED SOLUTION**

Scale factor = 15 − 5

Scale factor = 10

**A. EXPLAIN THE MISTAKE (1 POINT)**

A scale factor is a RATIO, found by dividing, not a difference: $\frac{15}{5} = 3$.

**B. SHOW CORRECT WORK (1 POINT)**

Scale factor = $\frac{\text{larger side}}{\text{smaller side}}$

Scale factor = $\frac{15}{5} = 3$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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## 9. Differences vs. ratios

**ORIGINAL PROBLEM**

Are triangles with sides 3, 4, 5 and 6, 7, 8 similar?

**INCORRECT WORKED SOLUTION**

Each side is 3 more than the matching side.

$6 = 3 + 3$, $7 = 4 + 3$, $8 = 5 + 3$

The triangles are similar.

**A. EXPLAIN THE MISTAKE (1 POINT)**

Similar triangles need equal RATIOS, not equal differences. $\frac{6}{3} = 2$, $\frac{7}{4} = 1.75$ and $\frac{8}{5} = 1.6$ are not equal.

**B. SHOW CORRECT WORK (1 POINT)**

Compare ratios of matching sides:

$\frac{6}{3} = 2$, $\frac{7}{4} = 1.75$, $\frac{8}{5} = 1.6$

The ratios are not equal, so the triangles are not similar.

## 10. Similar rectangles

**ORIGINAL PROBLEM**

Are a 4 by 6 rectangle and a 6 by 8 rectangle similar?

**INCORRECT WORKED SOLUTION**

Both are rectangles, so all angles are 90°.

The angles match, so the rectangles are similar.

**A. EXPLAIN THE MISTAKE (1 POINT)**

Similar polygons need matching angles AND proportional sides. Compare the sides: $\frac{6}{4} = \frac{3}{2}$ but $\frac{8}{6} = \frac{4}{3}$. The ratios are different.

**B. SHOW CORRECT WORK (1 POINT)**

Compare matching sides: $\frac{6}{4} = \frac{3}{2}$ and $\frac{8}{6} = \frac{4}{3}$

$\frac{3}{2} \neq \frac{4}{3}$

The rectangles are not similar.

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points